

Hypothetical causal assessment

If forced to name a single most probable explanation for the reported Ewing sarcoma pattern

Formal edition · 18 July 2026 · Companion to Preliminary Report v0.7 · Compiled from 68 graded sources · **Not a finding. Not an agency determination.**

Read this first. This document answers a counterfactual question: *if one had to commit to a single most probable explanation using only the assembled evidence, what would it be?* It is an exercise in **structured hypothesis ranking** — the reasoning an epidemiological consultant performs before any determination exists.

It does **not** establish causation. It does **not** accuse any organisation, contractor, manufacturer, agency or individual of causing any illness. The probabilities are **elicited subjective judgement**, not calculations. An official multi-agency review — Orange County Health Care Agency, California Cancer Registry, UC Irvine, and the County Agricultural Commissioner — is underway and has not reported; nothing here anticipates or substitutes for it. Health data is aggregate-only; no individual is identified.

68

GRADED SOURCES

79,473

PESTICIDE RECORDS

22

LITERATURE ENTRIES

13

ENVIRONMENTAL SITES

1 The question, decomposed

"What caused the cluster?" smuggles in two assumptions — that a cluster exists, and that it has an identifiable cause. A disciplined answer must address both. The question is therefore split:

1. **Is there a real excess?** Do registry-verifiable Ewing sarcoma diagnoses exceed the correctly age- and ancestry-adjusted expectation?
2. **If so, is it attributable?** Does any candidate survive both an *exposure-opportunity* test and a *disease-specificity* test?
3. **Which candidate is most probable?**

Public discussion collapses these into one. Keeping them apart is what makes a defensible conclusion possible.

1.1 Scoring criteria

CRITERION	WHAT IT ASKS
Exposure opportunity	Could children plausibly have contacted this at all?
Disease specificity	Any established or suggested link to <i>Ewing sarcoma</i> — not to cancer generally?
Temporal fit	Does the exposure window overlap the relevant birth cohorts?
Dose plausibility	Are concentration, persistence and bioavailability sufficient to matter?
Base-rate fit	How often has this <i>class</i> of explanation proved correct in comparable investigations?

The fifth criterion is the one usually omitted, and it carries the most weight. Cluster investigations have a long and consistent track record; ignoring it is the principal route by which motivated reasoning enters.

2 Population and baseline expectation

Ladera Ranch CDP (FIPS 0639114), American Community Survey 2020–2024:

MEASURE	VALUE	NOTE
Total population	23,793	26,170 at the 2020 Census; 22,980 in 2010
Children aged 0–19	9,115	38.3% of population — an unusually young community
Aged 10–19 (Ewing peak window)	4,906	Peak incidence falls at 10–15
Non-Hispanic white	63.6%	Materially raises the expected count
Asian / Hispanic / Black (NH)	13.6% / 12.9% / 0.9%	Lower-incidence ancestries
Median household income	\$184,458	68% hold a bachelor's degree or above
Person-years, ages 0–19, 2013–2026	127,610	Peak-age person-years: 68,684

2.1 Reference incidence rates

POPULATION	RATE	UNITS	SOURCE BASIS
US, all ages and races	2.93	per million/yr	SEER 1973–2004
US, ages 0–19 (central estimate)	3.0	per million/yr	SEER-based
North America, ages 10–19	4.58	per million/yr	CI5 2003–2012 peak
Southern Europe / Western Europe, 10–19	7.75 / 6.93	per million/yr	Spector 2021
East Asia, ages 10–19	1.59	per million/yr	Spector 2021
Ancestry IRR vs European	0.16	ratio	African; East Asian 0.30; Pacific Islander 2.18
All pediatric cancer, California 0–19	18.2	per 100,000/yr	CCR/SEER 2012–2016

Why ancestry governs this analysis. Ewing sarcoma incidence is roughly **nine times higher** in people of European ancestry than African ancestry — a differential driven by germline GGAA-microsatellite architecture, not environment. Applying an all-races rate to a 63.6% non-Hispanic-white population *understates the expected count and inflates every standardised incidence ratio downstream.*

3 Observed versus expected

Expected = person-years × rate. SIR = observed ÷ expected. Intervals are exact Poisson limits on the observed count.

SCENARIO	OBS.	RATE /M	EXPECTED	SIR	95% CI
S1 · Central 6 reported cases, ages 0–19, all-races SEER rate	6	3.00	0.383	15.7	5.8 – 34.1
S2 · Ancestry-adjusted rate raised for a 63.6% non-Hispanic-white population	6	4.00	0.510	11.8	4.3 – 25.6
S3 · Peak-age window ages 10–19 only, North American peak-age rate	6	4.58	0.315	19.1	7.0 – 41.5
S4 · Conservative count case definition tightened to 4	4	4.00	0.510	7.8	2.1 – 20.1
S5 · Outlier count the single-source '12 Ewing' figure	12	3.00	0.383	31.3	16.2 – 54.8
S6 · Leave-one-out one reported case removed as non-resident	5	4.00	0.510	9.8	3.2 – 22.9

Hypothetical — not a finding. Every row uses *unverified* public case reports and *estimated* population. California suppresses any rate built on fewer than 15 cases or a population under 10,000, so a place-level Ewing sarcoma rate is **statistically unpublishable**. Compare S4 with S5: the same community moves from SIR 7.8 to 31.3 purely on case definition.

3.1 The numerator problem

REPORTED COUNT	SCOPE AS STATED	SOURCE GRADE
"At least 6" Ewing sarcoma since 2013	Ladera Ranch only, Ewing only	B2 — local TV news
"About a dozen" rare cancers	Mixed cancer types across multiple cities	B2 — same outlet
"12 Ewing sarcoma"	Ladera Ranch, Ewing only	C — single low-reliability outlet

Only one of the reported Ewing cases carries both a published age and a diagnosis year. The widely repeated "dozen" explicitly aggregates *different cancers in different cities* — the textbook signature of ascertainment inflation rather than of a cluster. No count has been registry-verified, and the county's initial review, conducted with registry access, "did not find a particular pattern."

4 The evidence base

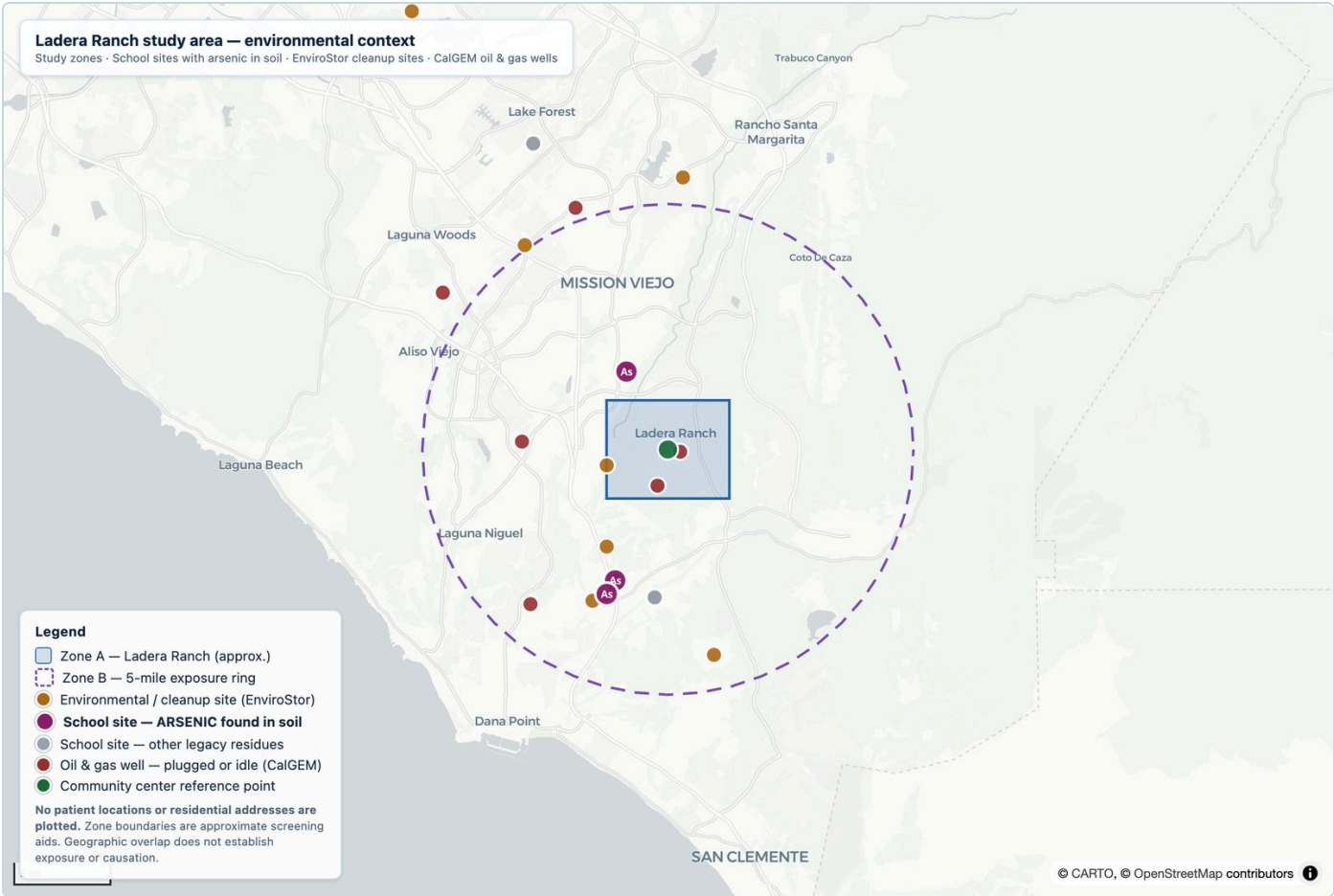


Figure 1. Study zones (approximate), DTSC EnviroStor cleanup sites, and CalGEM oil & gas wells. Two plugged exploratory wells lie within roughly a mile of the community centroid — a relationship that emerged only after correcting a 1.93-mile centroid error inherited from the original project brief. No patient locations are plotted. Basemap © CARTO, © OpenStreetMap contributors.

4.1 Environmental and cleanup sites, by corrected distance

SITE	CONTAMINANTS RECORDED	MILES
Blue Diamond Materials Plant	(unspecified)	1.3
Carl Hankey Elementary School	Arsenic, Lead, Methane	1.8
Meggitt (OC) Inc / former Endevco	Lead	2.3
Plant Depot School Site	Arsenic, Nitrate	2.9
Ambuehl Elementary School	DDD, DDE, DDT, Toxaphene	3.0
San Juan Elementary School	Arsenic, Chlordane, DDD, DDE, DDT, Lead	3.2
SOLAG Disposal Site	Arsenic, Cr, Lead, PAHs, Beryllium	3.5
Prima Deshecha Landfill	Landfill gas (methane), leachate; groundwater & LFG monitore	4.3
Alicia Towne Plaza	PCE	5.1
Polo Cleaners	1,1,1,2-tetrachloroethane; 1,1,2-TCA	5.6

Every legacy-pesticide record above is a **school** site. Not because schools received worse land, but because California Education Code §§17210 / 17213.1 mandates a Phase I assessment and DTSC review before a district acquires property — and **no equivalent mandate applies to residential subdivisions**. The absence of soil data for the residential footprint is a regulatory artifact, not evidence that the soil is clean.

4.2 Oil & gas wells (CalGEM)

WELL	STATUS	MILES
Citizens National Trust B-1 / Exxon Mobil	Plugged, Dry Hole	0.25
O'Neill #1 / Union Oil	Plugged, Dry Hole	0.77
Shumaker #1 / Conoco	Plugged, Dry Hole	2.98
Norswing & Halvorson #1	Idle, Oil & Gas	4.22
O'Neill Estate B-1 / Exxon Mobil	Plugged, Dry Hole	5.27
South Fullerton Oil Co. #1	Idle, Oil & Gas	5.59

Two plugged wells within one mile; three within 5 km; all six within 10 km — the exposure contrast used by the only published study reporting a Ewing-specific environmental association.

5 What is actually applied

California's 2023 Pesticide Use Report archive was downloaded and processed in full: **79,473 Orange County application records** across 382 chemicals and 45 site types.

SITE TYPE	RECORDS	POUNDS	LOCATED	% LOCATED
Structural pest control	55,442	403,983	0	0.0%
Landscape maintenance	15,383	110,664	22	0.1%
Nursery — outdoor containers	2,990	4,788	2,982	99.7%
Golf course turf	1,375	56,171	0	0.0%
Rights of way	1,130	34,192	48	4.2%
Agriculture (fruiting pepper)	183	13,411	183	100.0%

Overall, 94.6% of Orange County pesticide records carry no location at all. Landscape maintenance is reported — but 99.9% of those records lack township, range and section. Only agricultural and nursery categories are reliably geolocated. Compounding this, a federal cadastral query places Ladera Ranch in **unsectioned former land-grant territory** (T7S R7W / R8W, section 00). **The state's reporting system is structurally incapable of placing a pesticide application inside this community.**

5.1 Landscape-maintenance chemical mix, Orange County 2023

ACTIVE INGREDIENT	RECORDS	POUNDS
Glyphosate, isopropylamine salt	942	22,048.8
Diphacinone	834	1.8
Imidacloprid	599	1,098.4
Bifenthrin	558	799.9
Dicamba	520	95.9
Carfentrazone-ethyl	485	53.5
2,4-D, 2-ethylhexyl ester	479	1,315.4
Glyphosate, potassium salt	419	8,002.8
Triclopyr, butoxyethyl ester	362	1,329.4
Glufosinate-ammonium	336	10,177.1
Strychnine	331	32.0
Trinexapac-ethyl	316	596.0

Glufosinate in regional context. The ingredient named in Ladera Ranch application notices accounts for **442 county records and 10,532 lbs** in 2023, of which **336 records and 10,177 lbs** were landscape maintenance — 97% of its

poundage. Glyphosate remains larger across both salts (1,361 records, ~30,052 lbs). The community's herbicide programme is **unremarkable for the region**. Any causal account must explain why *here* and not in every comparable South Orange County community served by the same contractors and products.

6 Temporal analysis

Mass grading of the former agricultural land ran roughly **1999–2006**; reported diagnoses span **2013–2026**. A child could have been present during grading, including in utero, only if born on or before about 2007.

DIAGNOSIS YEAR	PEDIATRIC AGES WHOSE BIRTH COHORT OVERLAPS GRADING	SHARE
2013	14 of 15	93%
2016	11 of 15	73%
2019	8 of 15	53%
2022	5 of 15	33%
2024	3 of 15	20%
2026	1 of 15	7%

This separates two mechanisms that are routinely conflated. **Construction-era dust** is a time-bound event requiring presence during grading — substantially weakened for recent diagnoses, and unavailable for children born after ~2007. **Persistent soil residue** is a standing condition, unaffected by how long ago grading occurred. The single datable Ewing case (diagnosed August 2024, age ~17, implying birth between about August 2006 and August 2007) sits precisely on the boundary and discriminates nothing.

7 Quantitative context for the soil hypothesis

BENCHMARK	VALUE	SIGNIFICANCE
Southern California background arsenic (mean)	1.51 mg/kg	DTSC, n = 1,086 school-site samples
Background upper bound	12 mg/kg	The operative comparison for any local result
EPA residential screening level	0.68 mg/kg	<i>Below</i> virtually all natural background
California DTSC screening level	0.11 mg/kg	~100× below typical ambient concentrations
Washington State cleanup trigger	20 ppm	Formal former-orchard programme
Measured bioavailability, former orchard	0.31	Barber Orchard — total overstates dose ~3×
Child soil / dust ingestion, ages 1–6	50 / 60 mg/day	Regulators weight dust <i>above</i> soil

Exceeding a screening level in Orange County is normal, not evidence of contamination. Only a result above roughly 12 mg/kg would signify anything. At a comparable community, soil arsenic of 19.9 mg/kg produced **no significant correlation** with children's urinary arsenic ($r = 0.137$, $p = 0.39$); rice consumption was the significant predictor. Arsenic's established cancers are bladder, lung and skin — in adults, after decades.

8 Literature bearing on the question

STUDY	SIZE	EXPOSURE	EFFECT ESTIMATE	DIRECTION
Kendall 2020 (GB)	5,369 sarcoma cases	Paternal occupational	Textile dust OR 1.93 (1.01–3.68); no agriculture/agrochemical association	Weakens
Valery 2002 (AU)	106 cases / 344 controls	Parental farming	Farm residence OR 2.0 (1.0–3.9); farming father OR 3.5 (1.0–11.9)	Supports
Valery 2005 (pooled)	199 cases / 1,451 controls	Parental farming	Paternal OR 2.3 (1.3–4.1); maternal OR 3.9 (1.6–9.9)	Supports
Clark 2026 (CA)	558 cases / 27,800 controls	Oil & gas proximity	Active wells OR 0.88 (0.72–1.08); abandoned OR 1.27 (0.96–1.66)	Does not resolve
Clark 2025 (CA)	388 cases / 19,341 controls	Perinatal PM _{2.5}	Q4 gestation OR 0.87 (0.61–1.25) — null overall	Does not resolve
Vinson 2011 (meta)	40 studies	Parental pesticides	Leukaemia 1.48 (1.26–1.75); brain 1.49 (1.23–1.79); Ewing not reported	Supports (other cancers)
Spector 2021 (CI5)	15,874 cases	Ancestry / geography	Ages 10–19 per million: N. America 4.58; E. Asia 1.59; African IRR 0.16	Background
Bradman 2019	US cohorts	Residential mobility	55–58% of childhood-cancer cases moved between birth and diagnosis	Methodological

Ewing sarcoma has **no established environmental cause anywhere in the published literature**. It is defined by an EWSR1-FLI1 fusion (~85% of tumours) arising postnatally on a strongly ancestry-patterned germline background. The strongest positive signals are parental-farming *proxies* with confidence intervals touching 1.0; the largest and most rigorous occupational study found **no** agriculture or agrochemical association. No study links Ewing sarcoma to glyphosate, glufosinate, 2,4-D, or any specific pyrethroid.

8.1 Precedent from comparable investigations

INVESTIGATION	FINDING
CDC / CSTE guidance	~1,000 inquiries per year; investigations "unlikely to find an associated environmental contaminant"
Wake County, NC (Ewing sarcoma)	Observed did not exceed expected; no common exposure identified
Washington County, PA (Ewing sarcoma)	Local excess ~3× expected noted; not attributed to any environmental cause
McFarland, CA (mixed pediatric)	~3× expected; investigation unresolved ; four pesticides examined and dismissed

9 Hypothesis scoring

HYPOTHESIS	EXPOSURE	DISEASE FIT	TEMPORAL	DOSE	BASE RATE	P
Landscape herbicides	Moderate	None	Poor	Poor	Poor	5%
Legacy soil residue	Moderate	None	Good	Poor	Weak	7%
Abandoned oil & gas wells	Weak	Suggestive	Good	Unknown	Weak	8%
Chance + ascertainment	n/a	n/a	n/a	n/a	Strong	55%

9.1 Landscape herbicides

The dominant public hypothesis and, on evidence, the **weakest** environmental candidate. Exposure opportunity is genuine — children play on treated turf and residues track indoors. Everything else fails. No regulator classifies glufosinate as a carcinogen; glyphosate's IARC 2A concerns lymphoma; none is linked to Ewing sarcoma. Glufosinate's soil half-life is ~7.4 days, so exposure is pulsed rather than cumulative. Documented applications (2023–25) largely post-date the relevant windows for diagnoses in 2013–24. And the state's own data shows the practice is regionally ordinary.

9.2 Legacy agricultural soil residue

Stronger on exposure logic — arsenic never degrades, so it is a standing condition, and house dust, gardening and produce uptake survive surface cover. But background swamps the screening levels; measured bioavailability is ~0.31; lead arsenate was an *apple and pear* insecticide whereas this was citrus, barley and cattle land; mass grading likely diluted the plough layer; biomarker studies at comparable concentrations are null; and a systematic review found the childhood-cancer literature does not support an arsenic association.

9.3 Abandoned oil & gas well proximity

The **only candidate carrying a disease-matched published signal**: OR 1.27 (95% CI 0.96–1.66) for abandoned wells within 10 km, from a 558-case California study. The corrected geography places this community inside that contrast. Against it — the association is **not statistically significant**, the interval crosses 1, the effect was stronger in Hispanic children (12.9% of this population), the same study found *no* association with active wells, and these are mid-century plugged dry holes rather than production wells. Local groundwater is not used for supply.

9.4 Chance in a high-baseline-risk population

Not a fallback but a substantive hypothesis, usually dismissed without examination because it is unsatisfying. The base rate is decisive; the expected count is higher than public framing assumes because of ancestry; the numerator is inflated by construction; the boundary was drawn around the observed cases; and rare-disease small numbers are inherently unstable — moving from 6 cases to 4 halves the apparent ratio.

10 Conclusion

Most probable explanation

That there is no single environmental cause — that the apparent excess is most likely chance aggregation in a population with an elevated ancestry-specific baseline, amplified by media- and attorney-driven case ascertainment and a boundary drawn around the observed cases.

Forced to name one answer, that is it, and it is not a hedge. It carries the strongest base-rate support, aligns with the county's own initial review, and is the only explanation that does not require positing a mechanism never demonstrated for this cancer anywhere in the literature.

If the premise is granted — that a real, registry-confirmed excess exists — the most probable environmental contributor is proximity to abandoned oil and gas wells, solely because it is the only candidate with a Ewing-specific published association and the corrected geography places the community inside that exposure contrast. That signal is weak, non-significant, and held weakly.

The dominant public hypothesis — landscape herbicides — is the least probable of the environmental candidates.

10.1 Probability allocation

Elicited subjective judgement, not calculation. Sums to 100%.

EXPLANATION	P	
No single environmental cause — chance aggregation in a high-baseline-risk population, amplified by ascertainment and post-hoc boundary selection	55%	
A real excess whose cause is never identifiable	20%	
Real excess with a contribution from abandoned oil & gas well proximity	8%	
Real excess with a contribution from legacy agricultural soil residue	7%	
Real excess with a contribution from landscape herbicides	5%	
An exposure or mechanism not yet considered	5%	

11 What would change this conclusion

1. **Registry confirmation of counts and residency.** Six or more Ewing sarcoma diagnoses confirmed among children resident during 2013–2026 would drop the chance allocation from 55% to plausibly below 30%, making "real excess" the leading branch. *This single dataset moves more probability than everything else combined.*
2. **Birth years.** Cases clustered in births ≤ 2007 would revive construction-era dust; cases spread across birth years would eliminate it while leaving persistent residue intact.
3. **Soil and house-dust sampling.** Arsenic above ~ 12 mg/kg on the footprint, with the bioavailable fraction measured, would raise the soil branch materially.

4. **HOA and vendor application records.** Evidence of application intensity genuinely anomalous *for the region* — not merely present — would raise the herbicide branch.
5. **Well-integrity data.** Documented leakage or soil-gas anomalies at the two nearest plugged wells would strengthen the only disease-matched candidate.

12 What this document does not say

- It does **not** state that any illness was caused by any party, product or practice.
- It does **not** conclude that the families' concerns are unfounded. A low prior on a specific mechanism is a statement about evidence, not a verdict that nothing happened. These illnesses warrant investigation; that is why the official review exists.
- It does **not** anticipate that review. The agencies hold registry data this project cannot access, and their findings supersede this entirely.
- It is **not** evidence in any proceeding and should not be represented as such.

Ladera Environmental Health Research Platform · Hypothetical causal assessment, formal edition, generated 18 July 2026 from 68 graded sources (A1 20 · A2 18 · B1 9 · B2 8 · C 11 · D 2). Probabilities are elicited judgement, not calculation. An analytical exercise — not a finding, not an agency determination. Do not represent it as evidence of causation.